Core Language: RAII and Exceptions

RAII and scope guards, the exception-safety guarantees, `noexcept`, and choosing between exceptions and return values. Assumes Basics and OOP; move semantics, smart pointers, lambdas and types follow in their own lessons.

RAII: the central idea

RAII: tie a resource (memory, file, lock, socket) to an object. Acquire in the constructor, release in the destructor. Destructors run even when an exception is thrown, so nothing leaks.

```
class File {
    std::FILE* f_;
public:
    explicit File(const char* path) : f_(std::fopen(path, "r")) {}   // acquire
    ~File() { if (f_) std::fclose(f_); }                              // release
    File(const File&) = delete;                                       // one owner only
    File& operator=(const File&) = delete;
};
```

`vector`, `unique_ptr`, `lock_guard`, `fstream` are all RAII. Never manage a resource with a bare `new`/`delete`, `lock()`/`unlock()` or `fopen`/`fclose`: wrap it.

> KEY POINT RAII binds resource lifetime to object lifetime: automatic, exception-safe cleanup. It replaces `try/finally` and removes the need for a garbage collector.

Exceptions and stack unwinding

```cpp
double divide(double a, double b) {
    if (b == 0) throw std::invalid_argument("division by zero");
    return a / b;
}

try {
    divide(1, 0);
} catch (const std::invalid_argument& e) {        // catch by const reference
    std::cerr << e.what() << '\n';
} catch (const std::exception& e) {               // more general handler
goes AFTER specific ones
    std::cerr << "other: " << e.what() << '\n';
}
```

- **Throw by value, catch by `const&`** (catching by value slices a derived exception).
- **Stack unwinding**: leaving scopes runs destructors, in reverse order of construction, until a `catch` matches. This is why RAII is exception-safe.
- A **destructor must never throw**: if it throws while another exception is unwinding, `std::terminate` is called. Destructors are implicitly `noexcept`.
- A constructor that throws leaves no object: the destructors of already-built members and bases run, its own does not.
- `catch (...)` catches everything; rethrow with a bare `throw;` (never `throw e;`, which copies and slices).
- Derive your own exceptions from `std::exception` (or a standard subclass) so generic handlers still work.

> `COST` The no-throw path is almost free, but a throw is orders of magnitude slower than a return and its latency is unpredictable. Do not throw on a hot path or as ordinary control flow (see Exceptions, error codes and optional).

RAII everywhere: guards and standard wrappers

Almost every resource has a standard RAII wrapper. Use it instead of a manual acquire/release pair.

RESOURCE	RAII TYPE	RELEASED BY
heap memory	`vector`, `string`, `unique_ptr`	destructor frees the block
a file	`std::fstream`	destructor closes it
a mutex	`lock_guard`, `unique_lock`, `scoped_lock`	destructor unlocks
a thread	`std::jthread` (C++20)	destructor joins
anything else	your own **scope guard**	destructor runs a cleanup

```
template <typename F>
class ScopeExit {  // runs F when the scope ends, however it ends
    F f_;
    bool active_ = true;
public:
    explicit ScopeExit(F f) : f_(std::move(f)) {}
    ScopeExit(const ScopeExit&) = delete;  // a guard must not be copied
    void release() { active_ = false; }  // commit: cancel the cleanup
    ~ScopeExit() { if (active_) f_(); }
};

void transfer() {
    begin();
    ScopeExit rollback([] { undo(); });  // runs on return AND on an
exception
    step1(); step2();  // any of these may throw
    rollback.release();  // success: keep the changes
}
```

- A RAII class owns **one** resource. Copying it must be **deleted** (one owner) or a **deep copy**; moving transfers ownership (the Rule of Five, OOP lesson; Move Semantics).
- Prefer the **Rule of Zero**: hold resources in members that are already RAII, and you write no destructor at all.
- **Destructors must not throw** (see Exceptions and stack unwinding and Exception-safety guarantees).

KEY POINT "What is RAII?" The resource's lifetime is tied to an object: acquire in the constructor, release in the destructor. Destructors run on every exit path, including exceptions, so nothing leaks and no `finally` is needed.

Exception-safety guarantees and noexcept

LEVEL	PROMISE IF AN EXCEPTION ESCAPES	TYPICAL EXAMPLE
No-throw	it never throws	destructors, `swap`, moves, anything `noexcept`
Strong	state is exactly as before (commit or rollback)	`vector::push_back` when the element's move is `noexcept`
Basic	nothing leaks and invariants hold, but values may have changed	most operations

Getting the strong guarantee: build aside, then swap

```
void Table::replaceAll(const std::vector<Row>& fresh) {
    std::vector<Row> copy(fresh);  // may throw: *this is still untouched
    rows_.swap(copy);  // commit: swap cannot throw
}
```

Do everything that can fail on a **copy**, then publish the result with an operation that cannot throw. If anything throws before the swap, the object is unchanged.

noexcept

- `noexcept` promises a function will not throw. If it does anyway, the program calls `std::terminate`: there is no unwinding past it.
- Destructors are `noexcept` by default. A throwing destructor during unwinding means two exceptions at once, and the program terminates.
- **Mark move operations and `swap` `noexcept` when they cannot throw.** `std::vector` grows by moving its elements *only if* the move constructor is `noexcept`; otherwise it **copies**, to keep the strong guarantee. Compiled example: growing a vector whose elements have a `noexcept` move made 0 copies; with a possibly-throwing move it copied every element.
- `noexcept(expr)` asks the compiler whether an expression can throw; generic code uses it to pick the cheap path (`std::move_if_noexcept`).
- Do not mark a function `noexcept` just to be fast: a wrong promise turns an error into a crash.

Exceptions, error codes and optional

SITUATION	USE
a failure the caller cannot sensibly handle here (out of memory, broken invariant, constructor failure)	**exception**: it unwinds to code that can cope
an **expected** failure in normal flow (bad input, "not found", parse error)	return a value that says so: `std::optional`, an error code or enum, `std::expected` (C++23)
a hot path with steady latency (real-time code)	**no exceptions**: error codes or `optional` (Performance & Memory)
a `noexcept` context, or a destructor	never throw: catch inside and log or ignore

```cpp
std::optional<int> parseInt(std::string_view s) {  // failure is expected: no exception
    int v = 0;
    auto r = std::from_chars(s.data(), s.data() + s.size(), v);
    if (r.ec != std::errc{} || r.ptr != s.data() + s.size()) return std::nullopt;
    return v;
}

if (auto n = parseInt(input)) use(*n);  // the caller must look at the result
else report("not a number");
```

- **Exceptions** keep the normal path clean and cannot be silently ignored; the cost is that a throw is slow and the control flow is invisible.
- **Return values** make failure explicit and cheap; the cost is that every caller must check, and `[[nodiscard]]` helps remind them.
- Constructors cannot return a code: either throw, or use a factory function returning `optional` or an `expected`.
- Be consistent inside a module, and document what a function can throw.

KEY POINT "Exceptions or error codes?" Exceptions for rare failures that need unwinding, return values for expected failures and for hot paths. Mention RAII (so nothing leaks either way) and that a destructor must never throw.

Further reading by section

Where each section of this lesson is explained in more depth. Page numbers are PDF viewer pages.

RAII: the central idea

Programming: Principles and Practice (Stroustrup)

- 18.4 to 18.5 Resources and exceptions; resource-management pointers (RAII, unique_ptr) (PDF pages 542 to 553)

Exceptions and stack unwinding

C++ Primer (Lippman, Lajoie, Moo)

- 5.6 try blocks and exception handling (PDF pages 222 to 229)
- 18.1 Exception handling (PDF pages 801 to 813)

Programming: Principles and Practice (Stroustrup)

- 4.5 to 4.6 Run-time errors and exceptions (PDF pages 102 to 111)
- 18.4 to 18.5 Resources and exceptions; resource-management pointers (RAII, unique_ptr) (PDF pages 542 to 553)

Exception-safety guarantees and noexcept

C++ Primer (Lippman, Lajoie, Moo)

- 18.1 Exception handling (PDF pages 801 to 813)

Effective Modern C++ (Meyers)

- Item 14, Declare functions noexcept if they won't emit exceptions (PDF pages 108 to 115)

C++ Move Semantics (Josuttis)

- 7.1 and 7.4 Move constructors with and without noexcept; when and where to use noexcept (PDF pages 125 to 134 and 140 to 141)